



Associations of postpartum health with progesterone after insemination and endocrine signaling during early pregnancy in dairy cows

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ABSTRACT

Our objective was to investigate associations of postpartum health with serum progesterone (P4) concentrations after first artificial insemination (AI), expression of interferon-stimulated genes (ISG) in blood leukocytes, and serum pregnancy-associated glycoprotein (PAG) during early pregnancy in dairy cows. In this observational study, Holstein cows ($n = 475$) from 2 commercial herds were enrolled 3 wk before expected parturition. Body condition score (BCS) and lameness were assessed at enrollment, and total Ca, haptoglobin, and nonesterified fatty acids (NEFA) were measured in serum at 2 and 6 ± 2 DIM. Blood β -hydroxybutyrate and metritis were evaluated at $4, 8, 11$, and 15 ± 2 DIM, and purulent vaginal discharge (PVD) and endometritis based on endometrial cytology were diagnosed at 35 DIM. Onset of cyclicity was assessed by biweekly P4 measurements. BCS was assessed at enrollment and at 63 DIM, and lameness at enrollment and at 21 and 49 DIM. First AI was based on estrus detection until ~ 75 DIM, or synchronization for timed AI thereafter. Serum P4 was measured at d 8 and 12 after first AI, expression of ISG (*ISG15* and *RTP4*) in blood leukocytes was assessed at d 19, and PAG in serum of pregnant cows was measured at d 29, 33, and 40. Multivariable linear regression models were built including health variables (with metabolites categorized by ROC curve analysis) and covariates (season, milk yield, AI method, DIM), accounting for pregnancy status (for ISG and PAG models), repeated measures (for P4 and PAG models), and herd as random. Results are reported comparing cows affected vs. unaffected by each predictor. Serum P4 concentrations were lesser at d 8 after AI in cows that had total Ca ≤ 2.09 mM (3.6 vs. 4.0 ± 0.2 ng/mL) at 2 DIM or a clinical disease (3.7 vs. 4.0 ± 0.2 mg/mL), and P4 tended to be lesser at d 8 in cows that had PVD (3.6 vs. 4.0 ± 0.2 ng/mL) or any uterine disease (3.7 vs. 4.1 ± 0.2 ng/mL). Among cows pregnant at d

29, relative *ISG15* expression (fold change) at d 19 was greater in cows that had lameness [5.10 (4.18 to 6.18) vs. 3.14 (3.10 to 3.19)] but tended to be lesser in cows that had endometritis [3.20 (2.89 to 3.54) vs. 4.29 (4.20 to 4.35)], and relative expression of *RTP4* tended to be lesser in cows that had displaced abomasum [1.02 (0.62 to 1.67) vs. 1.75 (1.73 to 1.78)]. Serum PAG was lesser at d 29 [3.8 vs. 4.2 ± 0.2 sample-to-positive ratio (S:P)] in cows that had NEFA ≥ 0.73 mM at 6 DIM, and lesser at d 33 (3.9 vs. 4.5 ± 0.2 S:P) and 40 (3.1 vs. 3.7 ± 0.2 S:P) in cows that had prepartum BCS ≥ 3.75 . These findings suggest long-term effects of metabolic or reproductive tract disorders on luteal function after first AI, pregnancy recognition signaling, and placental function during early pregnancy. These associations may partially explain some of the pathways linking postpartum health and fertility in dairy cows.

Key words: Disease, Fertility, Reproduction, Transition health

INTRODUCTION

Transition dairy cows undergo several physiological changes associated with parturition and metabolic adaptations to the onset of lactation. These changes are commonly characterized by decreased feed intake, immune dysfunction, systemic and uterine inflammation, mineral imbalance, and negative energy balance (LeBlanc, 2012; Horst et al., 2021). When excessive or combined, these physiological stressors can become risk factors for metabolic or uterine disorders, observed in nearly half of early postpartum cows (LeBlanc, 2012; Bruinje et al., 2023b). A key consequence of metabolic and inflammatory diseases is impaired reproductive performance. Severe or prolonged hypocalcemia (Venjakob et al., 2018), elevated blood NEFA or haptoglobin (Hp), hyperketonemia (HYK), and uterine or non-uterine clinical diseases have been associated with reduced probability of estrus detection, pregnancy, and increased pregnancy losses (Ribeiro et al., 2016; Bruinje et al., 2023a; b). However, the mechanisms behind the associations between health disorders and reproductive performance are unclear.

Received August 9, 2023.

Accepted November 3, 2023.

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Following AI and ovulation, coordinated functions of the ovary, uterus, and early conceptus are crucial for successful establishment and maintenance of pregnancy. The increase in circulating P4 after formation of the corpus luteum (CL) is essential to support early embryo survival and development, conceptus elongation, and subsequent attachment (Carter et al., 2008; Forde et al., 2009; Clemente et al., 2009). Suboptimal P4 concentrations before d 7 after AI alter expression of endometrial genes associated with histotroph composition and the uterine environment and impair embryo development by d 14 (Forde et al., 2009, 2011). After d 16, pregnancy recognition (CL maintenance) is signaled by interferon-tau (IFN- τ) produced by the conceptus, which can stimulate the expression of endometrial genes associated with amino acid transport in vitro, and is likely associated with an enhanced uterine environment (Groebner et al., 2011). In pregnant cows, the binucleate giant cells at the fetal-maternal interface secrete PAG that can be detected in maternal circulation starting at 25 d of gestation (Giordano et al., 2012). Lower circulating pregnancy-specific protein B (PSPB) at 51 d of gestation was observed in high-risk pregnancies (small embryo size, slow heartbeat, or extra-amniotic membrane) compared with normal pregnancies (Peixoto et al., 2021). Lower PAG or PSPB at d 24 or 29 of gestation was reported in cows that lost the pregnancy by d 40 (Bruinje et al., 2023a) or by d 61 to 74 of gestation (Mercadante et al., 2016; Ribeiro et al., 2014; Stangaferro et al., 2021), and lower PAG was observed in pregnant cows that subsequently had calving problems compared with cows that had normal calving (Mercadante et al., 2016). These studies suggest that lesser PAG level during early gestation is associated with increased risk of unsuccessful pregnancy.

Although the physiological mechanisms above are crucial for early pregnancy establishment and maintenance, it is unclear if they are influenced by postpartum health. Expression of ISG in blood leukocytes (PBL) at d 20, and concentrations of PSPB from d 25 to 67 were reduced in cows with lesser P4 concentrations from 6 to 11 d after AI (Carvalho et al., 2017). When timed AI was used for first breeding, cows that had postpartum clinical diseases had lesser P4 concentration in plasma, impaired elongation of the preimplantation conceptus, and reduced ISG expression in PBL (Ribeiro et al., 2016). However, little is known about biological mechanisms linking postpartum health and metabolism with subsequent fertility in herds using mostly estrus detection for first breeding. We hypothesized that cows with postpartum health disorders would have lesser P4 concentrations in serum after first AI, lower relative expression of ISG in PBL, and lesser PAG levels in

serum during early pregnancy than unaffected cows. Therefore, our objective was to investigate associations of postpartum health with serum P4 concentrations after AI, expression of ISG in PBL, and serum PAG levels during early pregnancy in lactating Holstein cows inseminated mostly after estrus detection. A secondary objective was to assess relationships among P4 concentrations, expression of ISG, and PAG levels.

MATERIALS AND METHODS

This study was approved by the Animal Care Committee at the University of Guelph (AUP #4385) and is reported following the STROBE-Vet checklist guidelines (Strengthening the Reporting of Observational Studies in Epidemiology – Veterinary Extension; O'Connor et al., 2016).

Description of herds

Two commercial Holstein dairy herds in southwestern Ontario were enrolled in this prospective observational study. This study uses a subsample of cows in an observational study of transition period health and estrus detection that enrolled 1,532 prepartum Holstein cows (Bruinje et al., 2023b) between April 2019 and November 2020. The subsample [137 nulliparous (77 in herd A, 60 in herd B) and 337 parous cows (207 in herd A, 130 in herd B)] were also evaluated in a companion manuscript (Bruinje et al., 2023a), and was selected based on the weekday that first AI occurred, so sampling days after AI would be within the planned sampling interval described below. Herds were selected purposively, according to their proximity to the University of Guelph (Guelph, ON, Canada), their use of automated activity monitors (AAM) as a primary tool for reproductive management for first AI, enrollment in DHIA, established record keeping, and willingness to participate in the study. Descriptive information on the herds was obtained from DairyComp 305 (Valley Ag Software). Herd A milked approximately 470 Holstein cows 3 times a day using a parallel parlor system while herd B milked approximately 430 Holstein cows using an automated milking system with an average of 3.2 milkings per day. Production averaged 38 kg/d (11,558 kg 305 d milk at third DHIA milk test) for primiparous and 47 kg/d (14,256 kg 305 d milk) for multiparous cows in herd A, and 32 kg/d (9,931 kg 305 d milk) for primiparous and 41 kg/d (12,495 kg 305 d milk) for multiparous cows in herd B. During the study, 21-d pregnancy rate, 21-d insemination rate, and pregnancy risk averaged 25%, 62%, and 40% in herd A, and 23%, 63%, and 36% in herd B, respectively. Close-up dry

cows were housed in straw packs in both herds. Lactating cows were housed in freestall barns with sand beds (Herd A) or with waterbeds and shredded cardboard bedding (Herds B). Feed delivery occurred 2 to 3 times per day in herd A and once per day in herd B.

Assessment of postpartum health

Herds were visited and cows examined twice weekly by the research team, and the data collection scheme is presented in Figure 1. Cows were enrolled 21 ± 3 d before their expected parturition (257 ± 3 d of gestation). At enrollment, they were assessed for body condition score (BCS; 5-point scale with 0.25-point increments; Ferguson et al., 1994) and lameness (score ≥ 3 on a 1 to 5 scale; Sprecher et al., 1997). Cows were tested and had a negative result for bovine viral diarrhea at time of enrolment using a commercial ELISA kit (BVDV PI X2, IDEXX Laboratories Inc., Westbrook, ME). After calving, blood samples were taken through the middle coccygeal vein or artery at 2 ± 2 (1 to 4) and 6 ± 2 (4 to 7) DIM to measure serum concentrations of total Ca, Hp, and NEFA as described below. At 4, 8, 11, and 15 ± 2 DIM, β -hydroxybutyrate (BHB) concentrations were measured in blood using a hand-held device (Precision Xtra Meter and B-ketone test strips, Abbott Laboratories) and vaginal discharge was assessed using an intravaginal device (Metricheck, Simcro). Cows with a fetid, watery, and brownish or reddish discharge were classified as having metritis regardless of rectal temperature. Cows were checked for retained placenta (RP) during sampling, examinations, and routinely by the farm personnel.

At 35 ± 3 DIM, cows were examined for mucopurulent or purulent vaginal discharge [PVD; score 2 or 3 as in Sheldon et al. (2006)] and for endometritis (ENDO) using an intrauterine cytobrush technique. The endometrial cytology samples were collected from the dorsal aspect of the uterine body and smeared on glass microscope slides, stained with a modified Wright-Giemsa protocol (Hema 3 Stat Pack; Fisher Scientific), and examined under an optical microscope at $400 \times$ magnification. The proportion of polymorphonuclear cells (% PMN) of 300 total nucleated cells counted per slide was recorded. Serum P4 concentrations were measured in samples taken at 21, 35, 49, and 63 ± 3 DIM to estimate the onset of first ovulation when the first sample was ≥ 1.0 ng/mL. Cows were examined for lameness at 21 and 49 ± 3 DIM, and for BCS at 63 ± 3 DIM. All scores were assigned by the first or second author and the interobserver agreements (Cohen's weighted Kappa) were over 80%. Additional health data recorded by the farm staff were obtained from DairyComp including calving ease, RP, milk fever, displaced

abomasum, and mastitis (with a recorded treatment). Cows were treated according to each farm's predefined protocols. Clinical mastitis cases were treated with intramammary antibiotics, and cows with RP (herd A), RP with fever (herd B), or metritis were treated with injectable antibiotics. Cows with BHB ≥ 1.2 mM were administered oral propylene glycol daily for 3 to 5 d and injectable polyvitamins. Multiparous cows received 2 oral Ca boluses after calving, and cases of milk fever (difficulty standing or walking associated with hypocalcemia by 3 DIM) were treated with intravenous and subcutaneous Ca. Hoof trimming occurred at dry-off and at mid-lactation, and visibly lame cows identified by the staff were hoof trimmed or treated with injectable antibiotics (for foot rot).

Reproductive management for first insemination

First postpartum AI occurred primarily based on estrus detection by automated activity monitors. Herd A used a pedometer system (AfiAct, Afimilk, Kibutz Afikim) and, starting at 55 DIM, farm personnel checked for estrus alerts twice daily and inseminated cows within 12 h of the alert (AM or PM). Cows not detected in estrus by 73 ± 3 DIM were administered PGF_{2 α} (Bioestrovet, Vetoquinol or Lutalyse, Zoetis), and if not detected in estrus by 87 ± 3 DIM, entered a timed AI program [Ovsynch: GnRH (Fertiline, Vetoquinol) – 7 d – PGF_{2 α} – 56 h – GnRH – 12 h – AI]. Cows initiating the protocol without a CL received an intravaginal P4 releasing device (CIDR; Zoetis) from protocol d 0 to d 7. Herd B used a collar (SmartTag Neck, Nedap) for estrus detection and cows received AI in the AM within 24 h after an estrus alert, starting at 50 DIM. Primiparous cows not detected in estrus by 78 ± 3 DIM entered an Ovsynch program if a CL was present, or received GnRH (Fertiline, Vetoquinol) followed by Ovsynch 7 d later. Multiparous cows not detected in estrus by 60 ± 3 DIM were administered PGF_{2 α} and, if not detected in estrus 7 d later, entered either an Ovsynch (if a CL was present) or received GnRH followed by Ovsynch 7 d later (if no CL). In both herds, pregnancy diagnosis by transrectal ultrasound occurred once at approximately 33 ± 3 d after AI.

Sampling for P4, ISG, and PAG after insemination

Following first AI (study d 0), blood samples were collected at d 8 and 12 ± 1 for serum P4 measurement ($n = 475$ cows), at d 19 ± 1 for assessment of ISG expression in PBL ($n = 397$ cows), and at d 29, 33, and 40 ± 1 for measurement of serum PAG ($n = 469$). The sampling days for P4 measurements were chosen as they would represent key stages of early- and

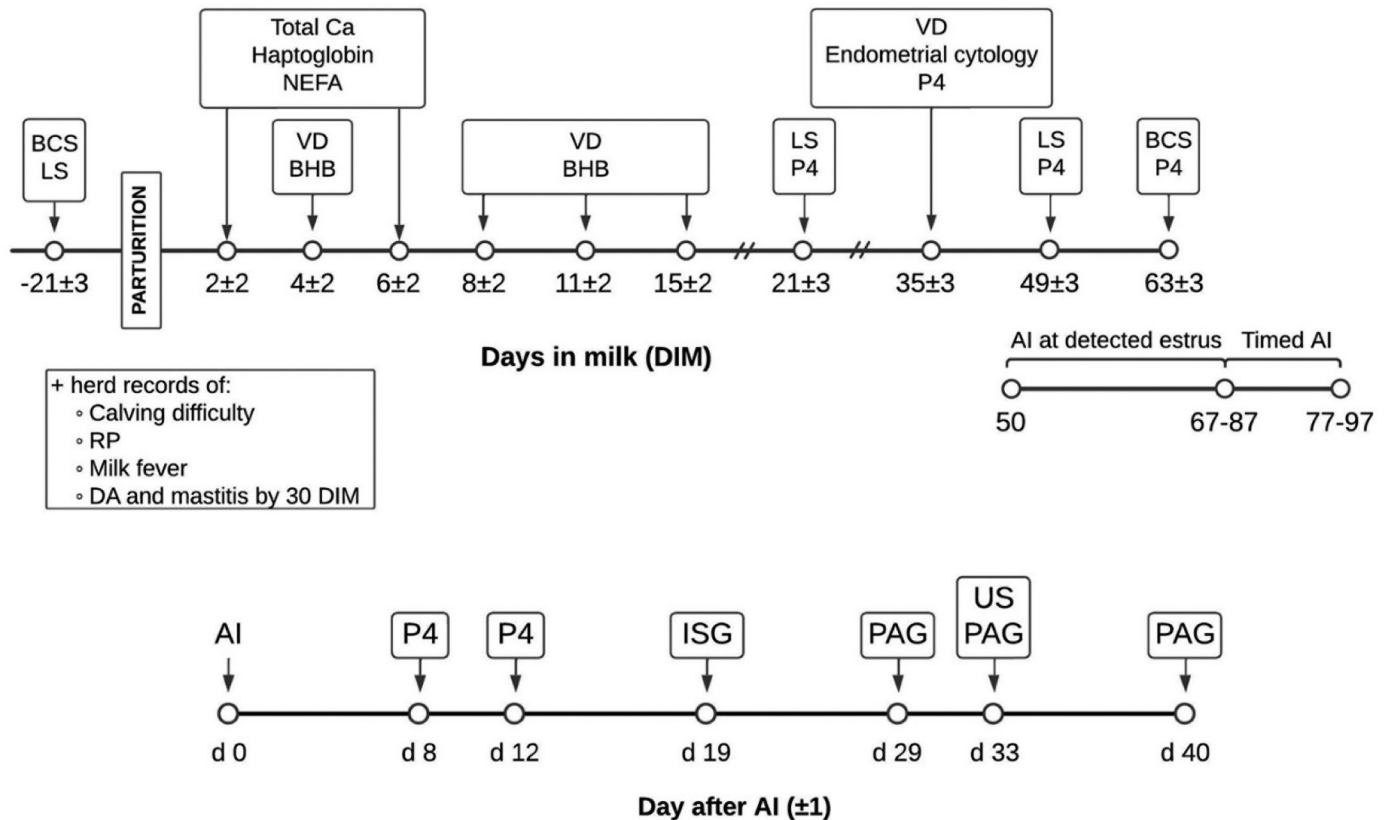


Figure 1. Schematic representation of data collection for assessment of postpartum health variables and outcomes after first artificial insemination (AI) in 475 Holstein cows in 2 commercial dairy herds in Ontario, Canada. Postpartum health variables were assessed based on examinations of body condition score (BCS), lameness score (LS), vaginal discharge (VD) score, endometrial cytology, and measurements of serum total Ca, haptoglobin, nonesterified fatty acids (NEFA), β -hydroxybutyrate (BHB), and progesterone (P4). First postpartum AI occurred either at estrus detection by activity monitors starting at 50 or 55 DIM or by timed AI (Ovsynch) starting between 67 and 87 DIM. Outcome variables following first AI were serum P4 concentrations ($n = 475$ cows), interferon-stimulated gene (ISG) expression in peripheral blood leukocytes ($n = 397$), and serum pregnancy-associated glycoprotein (PAG) levels in cows confirmed pregnant by ultrasound at $d 33 \pm 3$ (US; $n = 207$).

mid-diestrus when P4 is expected to be at intermediate and high levels, respectively (Molina-Coto et al., 2020). Similar time points for assessment of P4 and ISG expression were evaluated previously in association with postpartum health disorders (Molina-Coto et al., 2020; Ribeiro et al., 2016). Only cows with $P4 \geq 1.0$ ng/mL at d 8 and 12 after AI were included for the analysis of P4 concentrations, because lesser P4 concentrations possibly represented AI occurring at the wrong time relative to estrus or ovulation failure. Sampling days for PAG was based on a previous study reporting plasma PSPB concentrations substantially increasing in pregnant cows by d 28 after AI, followed by a peak between d 31 and 35, and a subtle decrease by d 42 in cows that subsequently lost the pregnancy compared with cows that maintained it (Ribeiro et al., 2014). Samples for PAG were stopped if cows returned to estrus or if they were diagnosed non-pregnant by transrectal ultrasound at 33 ± 3 d after AI, and only PAG samples from cows diagnosed pregnant by ultrasound at 33 d and that did

not return to estrus ($n = 207$ cows) were used for data analysis.

Laboratory analysis

Total Ca, Hp, NEFA, P4, and PAG in serum.

Blood samples were collected into 10-mL tubes without anticoagulant (Vacutainer Precision Glide, Becton Dickinson) and centrifuged at $1,500 \times g$ for 15 min within 4 h of collection. Serum was harvested and stored at -20°C until assayed at the Animal Health Laboratory (University of Guelph) for total Ca, Hp, and NEFA using an automated biochemistry analyzer (Cobas 6000 c 501, Roche Diagnostics). Control samples were tested and evaluated on a Levey-Kennings Quality Control chart before each assay. Total Ca was analyzed using the Cobas Calcium Gen.2 kit (Roche Diagnostics) with lower limit of quantification (LLOQ) of 0.63 mM and intra-assay coefficient of variation of 1.8%. The assay used for Hp was adapted from a previously described

method (Makimura and Suzuki, 1982; Skinner et al., 1991) and the LLOQ and intra-assay coefficient of variation were 0.02 g/L and 7.7%, respectively. Concentrations of NEFA were measured using the Randox Manual Rx Monza kit (Randox Laboratories), and the LLOQ and intra-assay coefficient of variation were 0.05 mM and 6.7%, respectively. A commercial ELISA kit was used to measure serum P4 (Ovucheck Plasma, Biovet) validated for use in serum (Broes and LeBlanc, 2014), and the LLOQ was 0.89 ng/mL and the inter- and intra-assay coefficients of variation were 6.7% and 8.9%, respectively. Pregnancy-associated glycoprotein was measured using an ELISA kit (Bovine Pregnancy Test Kit, IDEXX Laboratories Inc., Westbrook, ME) and the results were interpreted as the ratio between sample and positive control values (sample-to-positive ratio; S:P) according to the manufacturer's instructions. The inter- and intra-assay coefficients of variation were 1.7% and 0.9%, respectively.

Extraction of RNA and gene expression. Blood samples (3 mL) were drawn from the middle coccygeal vein or artery at $d 19 \pm 1$ after AI into tubes containing 6 mL of RNA stabilizing reagent (Tempus Blood RNA Tube, Applied Biosystems), immediately kept on ice, and stored at -20°C within 4 h of collection. The RNA was extracted using the Tempus Spin RNA Isolation Kit (Applied Biosystems) per the manufacturer's instructions, followed by a DNA removal step (RapidOut DNA Removal Kit, Thermo Scientific). On average, $82 (\pm 38)$ ng/ μL of RNA were isolated with ratio of concentration of nucleic acid to purified proteins (A260/A280) of 2:1. Samples were processed for quantitative real-time PCR at the Genomics Facility (Advanced Analysis Centre, University of Guelph) using a QuantStudio 7 Pro Real Time PCR system (Thermo Fisher Scientific Inc.). Briefly, 20- μL reactions comprised 1 μg RNA and 10 μL 2 $\times$ SsoAdvanced Universal Inhibitor-Tolerant SYBR Green Supermix (Bio-Rad), 0.8 μL of PCR forward and reverse primer mix at 5 μM (final primer concentration at 200 nM), 4.2 μL of water, and 5 μL of diluted cDNA. The cDNA was obtained from 1 μg RNA using a High-Capacity cDNA Reverse Transcription Kit (Applied BioSystems). The reactions were run in duplicate and included a polymerase activation step for 3 min at 98°C , followed by 40 cycles of a 2-step qPCR (10 s of 98°C denaturation, 30 s of 60°C combined annealing/extension). The expression of 4 genes was investigated (Table 1): interferon-stimulated gene 15 (*ISG15*) and receptor transporter gene 4 (*RTP4*) as target genes; β -actin (*ACTB*) and ribosomal protein L19 (*RPL19*) as reference genes. These genes were chosen as indirect measurements of conceptus development in lactating cows based on previous studies demonstrating greater expression of both genes in pregnant compared

with non-pregnant cows (Monteiro et al., 2014; Ribeiro et al., 2014, 2016). The difference between the target gene threshold cycle (C_T) and the geometric mean of the reference genes (Vandesompele et al., 2002) was used to obtain the ΔC_T values of *ISG15* and *RTP4*. Expression values were reported either on the additive inverse scale of ΔC_T ($\Delta C_T \times -1$) or as relative expression, for ease of interpretation. The relative expression was obtained by the difference in least squares means (LSM) between the comparison and the reference group. For each health variable evaluated, the reference group was non-pregnant cows based on PAG test at d 29 that were unaffected by the health disorder of interest for each comparison. The relative gene expression was then obtained by the PCR amplification efficiency ($E = 2$) raised to the power of the negative $\Delta\Delta C_T$ value (Livak and Schmittgen, 2001).

Statistical analyses

Data analyses were conducted using SAS (Studio 3.8 Edition; version 9.04; SAS Institute Inc.). Approximately 470 cows were targeted for P4 sampling, 400 cows for ISG expression, and 210 pregnant cows for PAG. This was based on sample size calculations performed a priori to detect differences with 80% power and 95% confidence using the POWER procedure, expecting a prevalence of at least one clinical disease of 45% and pregnancy risk of 40%. Detection of a difference in circulating P4 concentrations of approximately 2.8 vs. 3.2 ± 1.6 ng/mL at d 7 and 6.3 vs. 7.0 ± 1.6 ng/mL at d 15 in cow with vs. without a clinical disease (Ribeiro et al., 2016) required at least 250 cows per comparison group. A difference in relative expression of *ISG15* in pregnant cows with vs. without clinical disease of approximately 2.1 vs. 3.8 ± 4.0 was used as reference (Ribeiro et al., 2016), requiring at least 90 pregnant cows per group. For PAG, concentrations at d 32 of pregnancy in cows that subsequently lost or maintained that pregnancy (6.8 vs. 8.4 ± 4.0 ng/mL) were used as reference (Mercadante et al., 2016), and approximately 95 pregnant cows per group were required.

Descriptive statistics were obtained with the MEANS and FREQ procedures in SAS. Body condition score variables were categorized based on the median values: prepartum BCS was categorized as ≤ 3.50 or ≥ 3.75 , and the change between pre- and postpartum BCS (median = 0.50) was categorized as ≥ 0.50 loss, 0.25 loss, or no BCS loss (maintenance or gain in BCS). Milk yield data recorded at the first DHIA test at 38 (± 18) DIM were obtained from Lactanet and categorized as lowest, middle, or highest tercile for each parity group within each herd. Continuous values of metabolite concentrations were categorized based on cut points established

Table 1. Primer sequences for target (*ISG15* and *RTP4*) and reference genes (*ACTB* and *RPL19*) for quantification of mRNA expression in blood leukocytes 19 d after first insemination

Gene	NCBI sequence	Primer sequence
Interferon stimulated gene 15 (<i>ISG15</i>)	NM_174366	Forward: 5'-GGTATGAGCTGAAGCAGTT-3' Reverse: 5'-ACCTCCCTGCTGTCAAGGT-3'
Receptor transporter protein 4 (<i>RTP4</i>)	BC105539	Forward: 5'-CACATGTACCTGGAGAACCAGA-3' Reverse: 5'-AGGTAGCTCTGAAACCTTCCTG-3'
β -actin (<i>ACTB</i>)	AY141970	Forward: 5'-CTGGACTTCGAGCAGGAGAT-3' Reverse: 5'-GGATGTCGACGTCACACTTC-3'
Ribosomal protein L19 (<i>RPL19</i>)	NM_001040516	Forward: 5'-ATCGATCGCCACATGTATCA-3' Reverse: 5'-GCGTGCTTCCTTGGTCTTAG-3'

by receiver operating characteristic (**ROC**) curve analysis (LOGISTIC procedure in SAS) to predict pregnancy at first AI confirmed by ultrasound at d 33 in the larger data set ($n = 1,454$) that includes the data presented here (Bruinje et al., 2023b). The cut points and proportions of cows pregnant at first AI by category of each health variable in the present data set are presented in Table 2. Predictor variables evaluated as categories were parity, calving-related disorders (calving requiring assistance with a hard pull, stillbirth, or twin birth), milk fever, hypocalcemia, elevated Hp, elevated NEFA, HYK, RP, metritis, PVD, ENDO, DA, mastitis, lameness, BCS prepartum, BCS change, and days at first ovulation, in addition to the covariates of calving season, method for first AI (detected estrus vs. synchronization), categorized milk yield, and DIM at first AI as a continuous variable. The optimal cut points of ΔC_T values of *ISG15* and *RPT4* to predict pregnancy status (confirmed by PAG test on d 29) were described by ROC curve analysis. The agreement between *ISG15* and *RPT4* to classify pregnancy was calculated with Cohen's Kappa statistics (FREQ procedure in SAS), and Pearson correlation coefficients between *ISG15* and *RPT4* were obtained using the CORR procedure in SAS.

Data for P4 after AI, ISG, and PAG as outcomes were analyzed with mixed linear regression models using the GLIMMIX procedure in SAS. For P4 and PAG, first each variable was screened in individual models that also included the sampling day. Variables with $P \leq 0.20$ were offered to multivariable models that included health variables, parity, sampling day, and 2- and 3-way interactions of health variables with sampling day and parity, accounting for covariates (season, first AI method, milk yield, and DIM at first AI), repeated measures per cow, and farm as a random effect. The final multivariable models were selected using manual backward stepwise elimination retaining variables with $P \leq 0.10$. To check for confounders, removed variables were then individually re-entered and retained as confounders if their inclusion changed the coefficient of the significant predictors by $\geq 20\%$. With a similar approach, alterna-

tive P4 models were tested including only cows that received first AI at estrus detection.

Similarly, ISG data (ΔC_T of *ISG15* and *RTP4*) were analyzed by first screening each predictor variable in individual models including pregnancy status (based on PAG $S:P \geq 0.3$ at d 29) and their interaction. Variables with $P \leq 0.20$ were offered to multivariable models that included health variables, pregnancy status, parity, 2- and 3-way interactions of health variables with pregnancy and parity, covariates (as described above), and farm as a random effect. The final model selection was done as described above, except that interactions of predictor variables with pregnancy status were forced into the final models. Similarly, alternative models for ISG data were analyzed including either only cows that were classified as pregnant based on *ISG15* at d 19 (including true positives + false negatives), or those pregnant based on PAG at d 29.

Associations of P4 concentrations with ISG ΔC_T , of P4 with PAG, and of ISG ΔC_T with PAG, were evaluated in separate linear regression models with the GLIMMIX procedure. The models included the health variables and other covariates as described above, accounting for repeated measures per cow (for PAG models) and farm as a random effect.

For all outcomes, alternative multivariable models were tested with variables combined into uterine diseases (at least one of RP, metritis, ENDO, or PVD) or clinical diseases (at least one of milk fever, DA, mastitis, lameness, RP, metritis, or PVD). For all analyses, collinearity among predictors was checked by variance inflation factor. The distribution and homoscedasticity of standardized residuals were assessed graphically, and no transformations were required. The covariance structure (simple, compound symmetry, first-order auto regressive, first-order ante-dependence, Toeplitz, or unstructured) resulting in the lowest Akaike information criterion was used for models with repeated measures, and the SLICE command was used to assess partitioned LSM by sampling day or pregnancy status.

Table 2. Incidence of transition health disorders categorized based on blood markers or examined clinically in Holstein cows enrolled from two commercial dairy herds in Ontario, Canada

Variable	Threshold ¹	Affected / Total (n/n)	Prevalence or incidence (%)	P/AI (%) ² Affected	P/AI (%) ² Unaffected
Hypocalcemia ³					
2 ± 2 DIM	≤2.09 mM	262/473	55.4	43.1	45.0
6 ± 2 DIM	≤2.28 mM	278/464	59.9	43.2	46.2
Elevated Hp ³					
2 ± 2 DIM	≥1.54 g/L	79/473	16.7	31.7	46.5
6 ± 2 DIM	≥0.62 g/L	178/464	38.4	38.8	47.9
Elevated NEFA ³					
2 ± 2 DIM	≥0.73 mM	112/473	23.7	39.3	45.4
6 ± 2 DIM	≥0.63 mM	151/464	32.5	41.1	46.0
Hyperketonemia ³					
4 ± 2 DIM	≥0.8 mM	207/473	43.8	43.5	44.4
8 ± 2 DIM	≥0.9 mM	108/467	23.1	38.0	45.4
11 ± 2 DIM	≥0.7 mM	266/472	56.4	40.4	47.6
15 ± 2 DIM	≥0.7 mM	279/467	59.7	44.8	43.6
Retained placenta ⁴		37/475	7.8	37.8	44.3
Metritis ⁴		67/475	14.1	29.9	46.1
Endometritis ⁴	≥11.5% PMN	108/471	22.9	32.4	47.1
PVD ⁴		87/475	18.3	29.9	46.9
Lameness ⁵		88/475	18.6	41.6	44.3
Body condition score					
≥3.75 prepartum		179/475	37.7	45.3 ²	42.9
≥0.50 loss by 63 ± 3 DIM		247/474	52.1	41.7	46.3
Anovular ⁶ by 35 ± 3 DIM		147/475	30.9	44.2	43.6
Anovular ⁶ by 49 ± 3 DIM		60/475	12.6	35.0	45.1
Anovular ⁶ by 63 ± 3 DIM		25/472	5.3	36.0	44.3
Uterine disease ⁷		174/472	36.9	36.2	48.3
Clinical disease ⁸					
One		139/475	29.3	43.9	47.8
Multiple		66/475	13.9	27.3	—

¹Thresholds were selected based on the maximum combined sensitivity and specificity values to predict pregnancy at d 33 after first artificial insemination in a larger data set (n = 1,454 cows).

²Descriptive proportions of cows pregnant at first artificial insemination (P/AI) in the present data set for the affected and unaffected categories. Greater P/AI in categories of affected cows were due to unaccounted confounding variables such as parity and herd or by chance.

³Based on total Ca, haptoglobin (Hp), and nonesterified fatty acids (NEFA) concentrations measured in serum, and blood β-hydroxybutyrate.

⁴Retained placenta = failure to expel fetal membranes by 24 h after calving; Metritis = detection of fetid and colored (brownish/reddish) discharge at any of four examinations (4, 8, 11, and 15 ± 2 DIM); Endometritis = based on percentage of polymorphonuclear cells (PMN) assessed in endometrial cytology at 35 ± 3 DIM; PVD = mucopurulent or purulent vaginal discharge at 35 ± 3 DIM.

⁵Lameness: locomotion score ≥3 (1 to 5 scale) at 21 or 49 ± 3 DIM.

⁶Lack of ovulation based on serum progesterone <1.0 ng/mL measured on 21, 35, 49, and 63 ± 3 DIM.

⁷Diagnosed with at least one of retained placenta, metritis, endometritis, or PVD.

⁸Diagnosed with at least one of milk fever, retained placenta, metritis, PVD, displaced abomasum by 30 DIM, mastitis by 30 DIM, or lameness.

RESULTS

The proportion of nulliparous and parous cows enrolled in the study was 39 and 71%, respectively, and 61 and 28% of nulliparous and parous cows had prepartum BCS ≥3.75, respectively. Fifty-eight percent of cows calved in summer, 31% during fall, and 11% during winter, and first AI occurred in summer, fall, and winter in 11, 56, and 32% of cows, respectively. The prevalence of health disorders categorized based on blood markers or examined clinically are presented in Table 2. The median milk yield at first DHIA test was 38.2 kg/d in herd A and 31.1 kg/d in herd B in primiparous cows, and 57.1 kg/d in herd A and 44.8 kg/d in herd B in multiparous cows. First postpartum AI occurred on

average at 68 (±13) DIM and 91% of primiparous and 82% of multiparous cows received first postpartum AI after estrus detection on average at 66 (±9) DIM, with the remaining receiving timed AI at 87 (±15) DIM. Following AI, 91 and 95% of cows had elevated serum P4 (≥1.0 ng/mL) at d 8 and 12, respectively. Among cows pregnant at d 29 based on PAG, 99% had elevated P4 at d 8 and all had elevated P4 at d 12 after AI.

Progesterone after AI

The final multivariable model for P4 concentration included data from 410 cows, after excluding 13% of cows (62/475) with P4 < 1.0 ng/mL at d 8 or 12 after AI and 3 cows that had missing data on predictor

Table 3. Sensitivity (Sn), specificity (Sp), false negative fraction (FNF), positive and negative predictive values (PPV and NPV), accuracy, apparent prevalence (AP; classified as pregnant based on *ISG15* or *RTP4*) and true prevalence (AP + FNF) of the optimal cut points of *ISG15* and *RTP4* ΔC_T , determined by receiver operating characteristic curve analysis, to classify pregnancy

Interferon-stimulated gene ¹	Sn	Sp	FNF	PPV	NPV	Accuracy	AP	TP
<i>ISG15</i>	88.2% (164/186)	69.9% (137/196)	31.1% (59/196)	73.5% (164/223)	86.2% (137/159)	78.8% (301/382)	58.4% (223/382)	64.1% (245/382)
<i>RTP4</i>	72.0% (134/186)	65.8% (129/196)	34.2% (67/196)	66.7% (134/201)	71.3% (129/181)	68.8% (263/382)	52.6% (201/382)	66.2% (253/382)

¹Expression of *ISG15* and *RTP4* were assessed in peripheral blood leukocytes on d 19 $\pm$ 1 after first artificial insemination, and pregnancy status was determined by pregnancy-associated glycoprotein test on d 29 $\pm$ 1 after insemination.

variables. As presented in Table 4, variables associated with lesser P4 concentration at d 8 were hypocalcemia at 2 $\pm$ 2 DIM ($P = 0.02$), PVD ($P = 0.09$), or the occurrence of a clinical disease ($P = 0.04$) or a uterine disease ($P = 0.04$). Variables associated with lesser P4 concentration at d 12 were HYK at 5 $\pm$ 2 DIM ($P = 0.01$) or the occurrence of a clinical disease ($P = 0.08$). Conversely, cows with Hp ≥ 1.54 g/L at 2 $\pm$ 2 DIM had greater concentrations of P4 at d 8 and 12 than cows with lower Hp concentration. The proportion of cows receiving first AI after timed AI, due to not being detected in estrus, was 14 and 11% in cows with and without hypocalcemia, 13% in cows either with or without elevated Hp, 10 and 15% in cows with and without HYK, 14 and 13% in cows with and without PVD, and 16 and 11% in cows with and without a clinical disease, respectively.

In models including only cows receiving first AI at detected estrus, the risk factors for reduced P4 concentrations were similar, but did not retain hypocalcemia and uterine disease ($P > 0.10$) and included cyclicity status at 21 DIM ($P = 0.07$). Cows that were anovular (P4 < 1.0 ng/mL) at 21 DIM tended to have lower P4 concentration at d 12 compared with cows that were cyclic (P4 ≥ 1.0 ng/mL) at 21 DIM (Table 4).

Expression of ISG in PBL

For evaluation of ISG, 15 cows were not used in the analyses due to low estimated RNA concentration (A260/A280) or abnormal C_T values of reference genes. The test characteristics of using the optimal cut points of *ISG15* and *RTP4* ΔC_T to classify pregnancy status are presented in Table 3, and *ISG15* had superior accuracy in classifying pregnancy status. The Cohen's kappa statistics for agreement between *ISG15* and *RTP4* was moderate ($\kappa = 0.41$; 95% confidence interval 0.33 to 0.51), and the correlation between *ISG15* and *RTP4* was moderate in either open ($r = 0.44$; $P < 0.001$) or pregnant ($r = 0.41$; $P < 0.001$) cows at d 29.

The final multivariable linear regression models for *ISG15* and *RTP4* excluded data on 5 and 10 cows, respectively, due to missing data on predictor variables.

Variables associated with expression of *ISG15* in the final multivariable model included pregnancy status ($P < 0.001$), HYK at 11 $\pm$ 2 DIM ($P = 0.03$), ENDO ($P < 0.01$), an interaction between lameness and pregnancy status ($P = 0.06$), and an interaction between clinical disease and pregnancy status ($P = 0.04$) (Figure 2). Because of the observed associations of lameness and clinical disease with *ISG15* expression in pregnant cows, we built a multivariable model that included a variable categorized into lameness only ($n = 48$), other clinical disease only ($n = 94$), both lameness and other clinical disease ($n = 21$), or neither ($n = 214$), and *ISG15* expression was similar among groups ($P = 0.31$) with no interaction between group and pregnancy status ($P = 0.11$).

Variables associated with *RTP4* expression were pregnancy status ($P < 0.001$), parity ($P = 0.07$), HYK at 8 $\pm$ 2 DIM ($P = 0.10$), and DA ($P < 0.001$), with calving season and milk yield accounted in the model as covariates (Figure 3). There was no association of clinical disease with *RTP4* expression. Relative expression values (ratios and 95% CI) for the results presented in Figures 2 and 3 respective to the multivariable models of *ISG15* and *RTP4* are presented in detail in Supplemental Table S1.

In models that either included only cows classified pregnant based on *ISG15* at d 19 or cows pregnant based on PAG test at d 29 (and excluded open cows), the relative *ISG15* expression tended to be lesser in cows classified pregnant at d 19 that had a calving-related disorder or HYK 11 $\pm$ 2 DIM (Figure 4a), while it tended to lesser in cows classified as pregnant at d 29 that had endometritis and it was greater in cows that had lameness or a clinical disease (Figure 4b). Relative expression of *RTP4* at d 19 was greater in primiparous than multiparous cows [1.96 (1.84 to 2.08) vs. 1.48 (1.43 to 1.53); $P < 0.01$] and tended to be lesser in the 5 pregnant cows that had DA [1.02 (0.62 to 1.67) vs. 1.75 (1.73 to 1.78); $P = 0.09$] compared with 185 pregnant cows without DA. There was no association between health variables and expression of *RTP4* in models including only cows classified as pregnant at d 19.

Table 4. Results of multivariable linear regression models of associations between postpartum health variables and serum progesterone concentrations (least squares means $\pm$ SEM; ng/mL) at d 8 and 12 after first postpartum artificial insemination (AI) in lactating Holstein cows from two commercial herds

Risk factor	Day 8		Day 12		<i>P</i> -value		
	Unaffected	Affected	Unaffected	Affected	Risk factor	Day	Factor by Day Interaction
All cows (n = 410) ¹							
Hypocalcemia ²	4.0 $\pm$ 0.2 ^a	3.6 $\pm$ 0.2 ^b	5.6 $\pm$ 0.2	5.5 $\pm$ 0.2	0.19	<0.001	0.04
Elevated Hp ³	3.5 $\pm$ 0.2 ^a	4.0 $\pm$ 0.2 ^b	5.3 $\pm$ 0.2 ^c	5.8 $\pm$ 0.3 ^d	0.03	<0.001	0.03
Hyperketonemia ⁴	3.9 $\pm$ 0.2	3.7 $\pm$ 0.2	5.8 $\pm$ 0.2 ^a	5.3 $\pm$ 0.2 ^b	0.06	<0.001	0.83
PVD ⁵	4.0 $\pm$ 0.1 ^c	3.6 $\pm$ 0.2 ^d	5.8 $\pm$ 0.2	5.3 $\pm$ 0.3	0.07	<0.001	0.80
Clinical disease ⁶	4.1 $\pm$ 0.2 ^a	3.7 $\pm$ 0.2 ^b	5.9 $\pm$ 0.2 ^c	5.5 $\pm$ 0.2 ^d	0.03	<0.001	1.00
Uterine disease ⁷	4.1 $\pm$ 0.2 ^a	3.7 $\pm$ 0.2 ^b	5.9 $\pm$ 0.2	5.6 $\pm$ 0.2	0.07	<0.001	0.57
Bred at ED (n = 357) ¹							
Elevated Hp ³	3.7 $\pm$ 0.2 ^a	4.2 $\pm$ 0.2 ^b	5.4 $\pm$ 0.2 ^a	6.0 $\pm$ 0.3 ^b	0.01	<0.001	0.66
Hyperketonemia ⁴	4.0 $\pm$ 0.2	3.9 $\pm$ 0.2	5.9 $\pm$ 0.2 ^a	5.5 $\pm$ 0.2 ^b	0.15	<0.001	0.02
PVD ⁵	4.2 $\pm$ 0.2	3.8 $\pm$ 0.2	5.9 $\pm$ 0.2 ^c	5.4 $\pm$ 0.3 ^d	0.07	<0.001	0.68
Anovular at 21 DIM ⁸	4.1 $\pm$ 0.2	3.8 $\pm$ 0.2	5.9 $\pm$ 0.3 ^c	5.5 $\pm$ 0.2 ^d	0.07	<0.001	0.72
Clinical disease ⁶	4.2 $\pm$ 0.2 ^c	3.8 $\pm$ 0.2 ^d	5.9 $\pm$ 0.2	5.5 $\pm$ 0.2	0.06	<0.001	0.92

^{a,b} $P \leq 0.05$; ^{c,d} $0.05 \leq P \leq 0.10$ (within sampling day for each risk factor).

¹Models excluded cows with serum P4 < 1.0 ng/mL at d 8 or 12 (n = 56) and included either all cows (bred at estrus detection (ED) or timed AI) or only cows bred at ED, accounting for parity or season at AI as a covariate and herd as a random effect.

²Total Ca concentration ≤ 2.09 mM in serum at 2 ± 2 DIM.

³Haptoglobin (Hp) concentration ≥ 1.54 g/L in serum at 2 ± 2 DIM.

⁴ β -hydroxybutyrate concentration ≥ 0.8 mM in blood at 4 ± 2 DIM.

⁵Mucopurulent or purulent vaginal discharge (PVD) at 35 ± 3 DIM.

⁶Clinical disease = at least one of milk fever, retained placenta, metritis, or displaced abomasum, mastitis, purulent vaginal discharge, or lameness, evaluated in a separate multivariable model.

⁷Uterine disease = at least one of retained placenta, metritis, purulent vaginal discharge, or endometritis ($\geq 11\%$ polymorphonuclear cells in endometrial cytology at 35 ± 3 DIM), evaluated in a separate multivariable model.

⁸Serum progesterone concentration <1.0 ng/mL at 21 ± 3 DIM.

There was an interaction among P4 concentration at d 8 after AI, parity, and pregnancy outcome in their association with expression of *RPT4* ($P = 0.09$). As presented in Figure 5, in multiparous cows, there were weak associations of P4 concentration with *RTP4* expression. A 1 ng/mL increase in P4 concentration at d 8 was associated with 0.11 greater ($P = 0.05$) *RTP4* expression ($\Delta C_T \times -1$) in open cows and with 0.10 greater ($P = 0.09$) *RPT4* expression in pregnant cows). There was no association between P4 concentration and *RTP4* in primiparous cows.

Pregnancy-associated glycoprotein

Of cows sampled for PAG, 54% (252/469), 50% (236/469), and 45% (209/469) were pregnant based on PAG S:P > 0.3 at d 29, 33, and 40 after AI, respectively. At d 33, one cow was classified open by PAG but pregnant by ultrasound. In the multivariable model, 5 cows were excluded due to missing data on predictor variables. The proportions of cows diagnosed open by ultrasound at d 33 but pregnant by PAG at d 29, 33, and 40 were 13% (31/232), 8% (18/232), and 2% (4/232), respectively. The S:P in cows classified pregnant based on ultrasound at d 33 averaged (LSM $\pm$

SEM) 4.1 ± 0.1 at d 29, increasing to 4.3 ± 0.1 at d 33 ($P < 0.001$), and decreasing to 3.5 ± 0.1 at d 40 after AI ($P < 0.001$). Overall PAG S:P was greater in primiparous than in multiparous cows ($P = 0.08$) and in cows with elevated prepartum BCS (≥ 3.0 vs. ≤ 3.50 ; $P < 0.01$), and there was an interaction between NEFA at 6 ± 2 DIM and sampling day ($P = 0.05$) (Figure 6). The proportion of primiparous cows with prepartum BCS ≥ 3.75 and NEFA ≥ 0.73 mM was 33% and 55%, respectively, compared with 26% and 28% in multiparous cows. Uterine disease or clinical disease were not associated with PAG levels.

As presented in Figure 7, there was an interaction between *RPT4* expression and parity in their association with PAG level on d 29 ($P = 0.07$) and on d 40 ($P = 0.03$), but *RPT4* was not associated with PAG on d 33. In primiparous cows, one-unit increase in *RTP4* expression ($\Delta C_T \times -1$) was associated with 0.43 greater PAG S:P on d 29 ($P = 0.03$) and 0.59 greater PAG S:P on d 40 ($P < 0.01$). There was no association between expression of *RPT4* and PAG levels in multiparous cows.

There was an association between *ISG15* expression and PAG S:P on d 29 ($P = 0.04$), and an interaction between *ISG15* expression and parity in their asso-

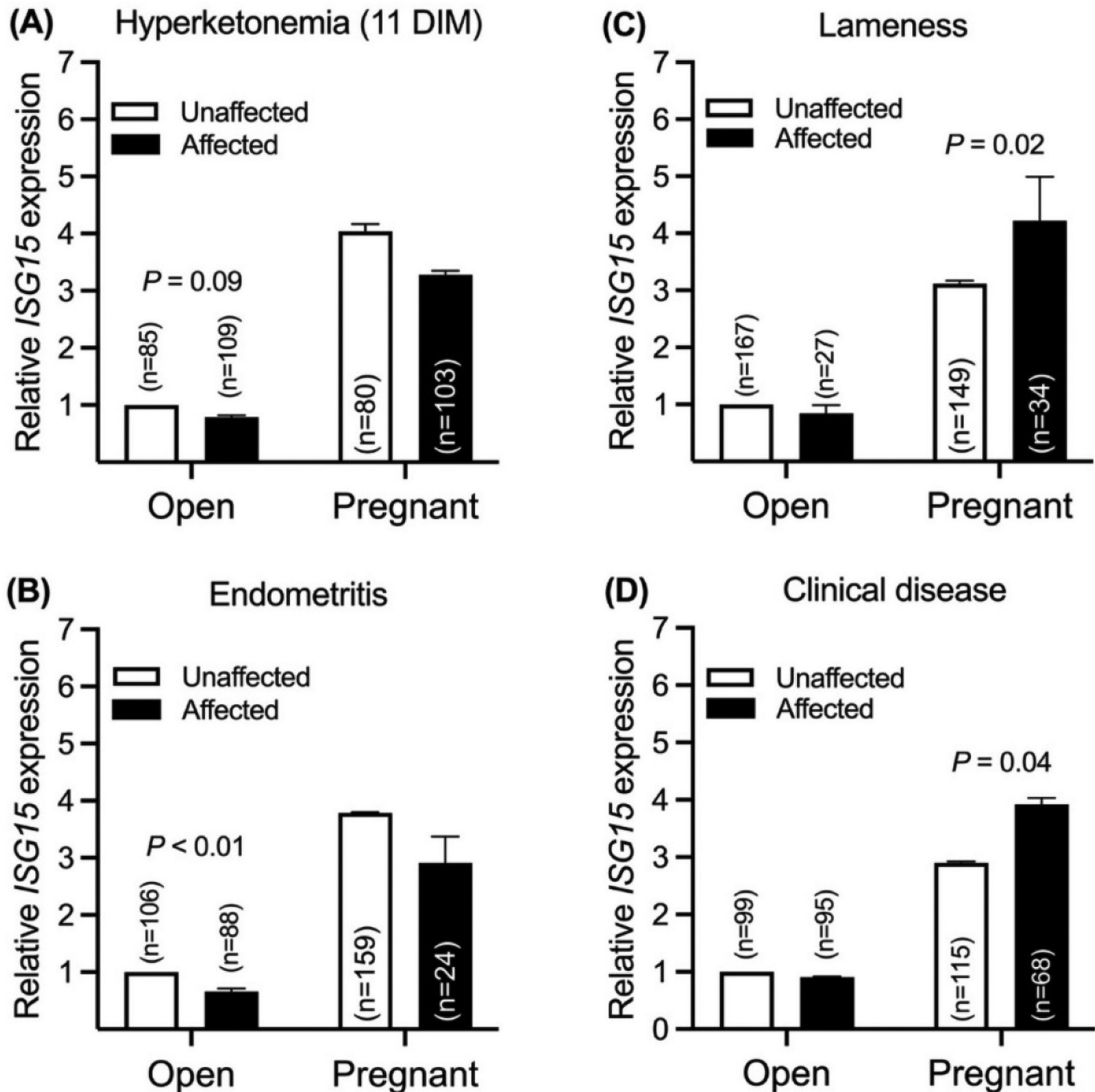


Figure 2. Health variables included in a multivariable model of relative expression of *ISG15* ($2^{\Delta\Delta CT}$) in blood leukocytes at d 19 after first insemination relative to unaffected cows (for each panel) subsequently diagnosed open based on PAG test at d 29 in Holstein cows from 2 herds. Therefore, relative expression for open, unaffected cows was 1. Results are from multivariable models accounting for herd as a random effect. Ratio values are presented in Supplemental Table S1. A) Hyperketonemia = blood β -hydroxybutyrate ≥ 0.7 mM at 11 ± 2 DIM, B) Endometritis = $\geq 11.5\%$ polymorphonuclear cells in endometrial cytology at 35 ± 3 DIM, C) Lameness at 49 ± 3 DIM; D) Clinical disease = any of retained placenta, metritis, purulent vaginal discharge, milk fever, displaced abomasum, mastitis, or lameness.

ciation with PAG S:P on d 40 ($P < 0.01$) (Figure 7). Expression of *ISG15* was not associated with PAG S:P on d 33. Each one unit increase in *ISG15* ($\Delta C_T \times -1$)

was associated with 0.14 greater PAG S:P on d 29 ($P = 0.04$). In primiparous cows, each one unit increased in *ISG15* was associated with 0.50 greater PAG S:P on d

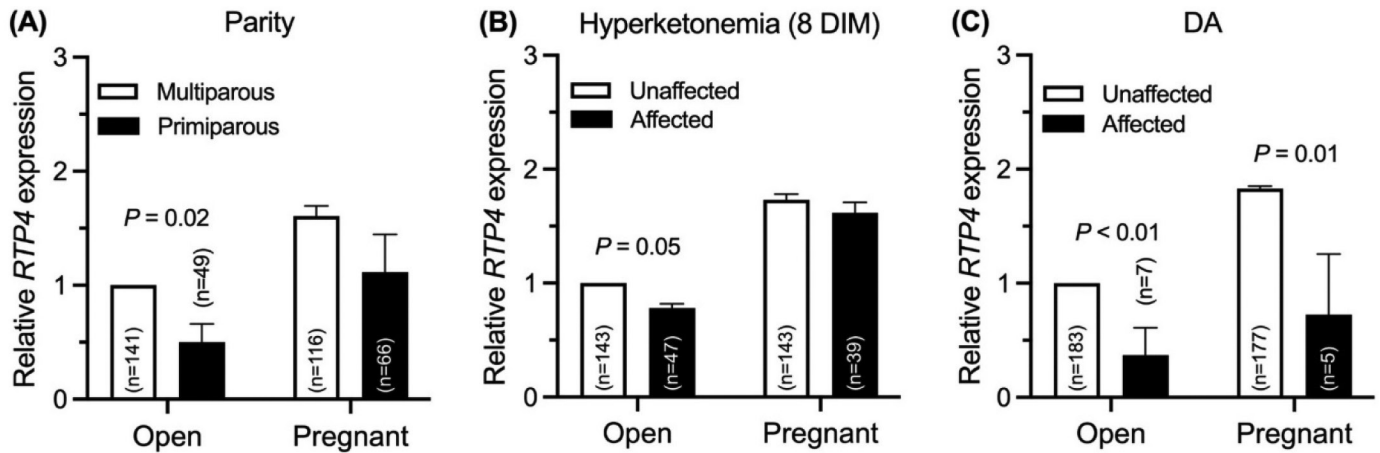


Figure 3. Variables associated with relative expression of *RTP4* ($2^{-\Delta\Delta CT}$) in blood leukocytes at d 19 after first insemination relative to unaffected cows (for each panel) subsequently diagnosed open based on PAG test at d 29, in Holstein cows from 2 herds. Therefore, relative expression for open, unaffected cows was 1. Ratio values are presented in Supplemental Table S1. Results are from a multivariable model accounting for calving season and milk yield at first DHIA test as covariates and herd as a random effect. A) Primiparous or multiparous cows; B) Hyperketonemia = blood β -hydroxybutyrate ≥ 0.9 mM at 8 ± 2 DIM; C) Displaced abomasum (DA).

40 ($P < 0.01$). There was no association between *ISG15* expression and PAG levels in multiparous cows.

DISCUSSION

We evaluated associations of health disorders during the early postpartum period with serum P4 concentrations after first AI, expression of ISG in PBL, and PAG levels during early pregnancy. Hypocalcemia at 2 DIM, HYK at 5 DIM, PVD, and the occurrence of a clinical or a uterine disease were associated with lower circulating P4, while elevated Hp at 2 DIM was associated with greater P4 concentrations. In models including both pregnant and open cows based on PAG at d 29 after AI, *ISG15* expression was greater in pregnant cows that had lameness or at least one clinical disease compared with pregnant cows without these risk factors, and *RTP4* expression was lesser in pregnant cows that had a DA than pregnant cows that did not have DA. Considering only pregnant cows based on *ISG15* at d 19 or PAG at d 29, calving problems, HYK at 11 DIM, or ENDO were associated reduced *ISG15* expression, and DA was associated with reduced *RTP4* expression. Conversely, lameness at 49 DIM was associated with greater *ISG15* expression. During early pregnancy, PAG levels were lower in cows that had high BCS prepartum or that had elevated NEFA postpartum.

There are few reports evaluating associations between metabolic and inflammatory markers and subsequent luteal function. Effects of hypocalcemia on re-establishment of cyclicity postpartum have been observed. Cows with low blood Ca either immediately after calving (Jeong et al., 2018), during the first 3 DIM (Caixeta et

al., 2017), or at 7 or 14 DIM (Ribeiro et al., 2013), were less likely to resume cyclicity by 49 to 56 DIM than normocalcemic cows. Factors influencing delayed cyclicity could be similarly associated with impaired luteal function after first AI (and reduced P4 concentrations), although our analysis of P4 concentrations included only cows that had indication of a functional CL ($P4 \geq 1.0$ ng/mL) after AI. In the model that excluded cows bred after timed AI, which occurred only if cows were not detected in estrus, hypocalcemia was not retained in the final model. This could have occurred because a slightly greater proportion of cows with hypocalcemia were inseminated after timed AI (14%) compared with cows without hypocalcemia (11%), therefore having lower P4 concentrations than cows bred at detected estrus. Hypocalcemia is also associated with the risk of health disorders such as HYK, metritis, or DA (McArt and Neves, 2020) and hypocalcemia at 4 DIM is associated with reduced odds of pregnancy at first AI and longer time to pregnancy (Seely and McArt, 2023). It could be a marker of feed intake or of systemic inflammation (Neves, 2023). Therefore, it is plausible that cows with hypocalcemia have impaired luteal function and lower circulating P4 during early and mid-di estrus after first AI, although the effect may be mediated by intervening pathways.

We expected that, as a marker of systemic inflammation, elevated Hp would be unfavorably associated with indicators of fertility. However, elevated Hp at 2 DIM was associated with somewhat greater P4 concentrations after AI. Circulating Hp at 3 and 6 DIM was greater in cows that subsequently developed metritis (Huzzey et al., 2009) and in cows with impaired follicu-

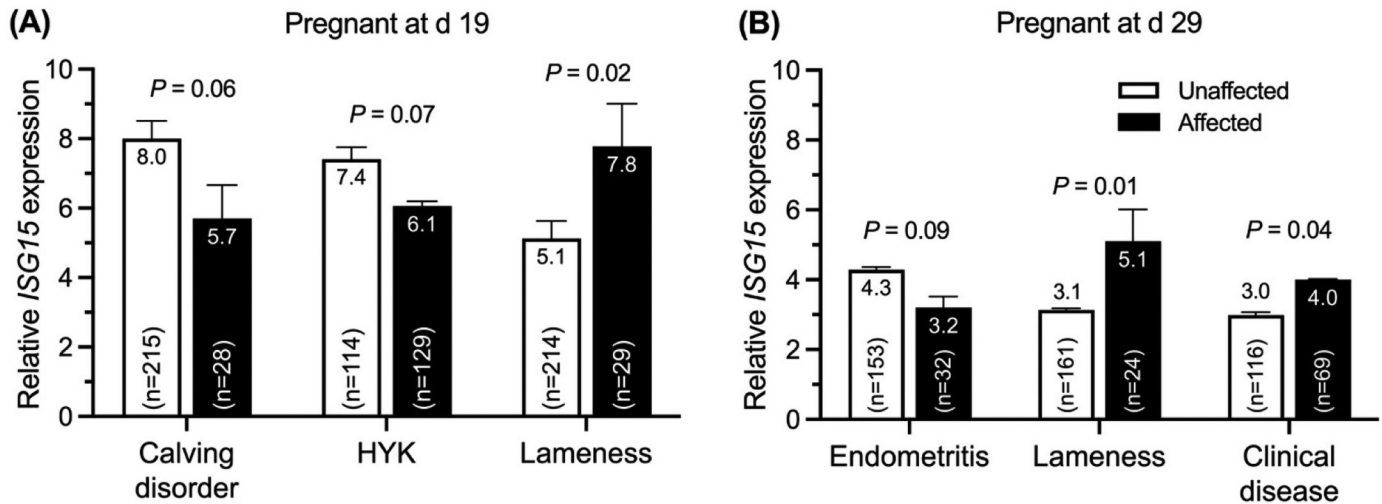


Figure 4. Variables associated with relative expression of *ISG15* ($2^{-\Delta\Delta CT}$) in peripheral blood leukocytes at d 19 after first artificial insemination (AI) relative to unaffected cows classified open by the same test at the same time (not shown). Results are from multivariable models that included only cows classified as pregnant by *ISG15* at d 19 (panel A) or PAG test at d 29 (panel B), accounting for AI method (estrus detection or timed AI) as a covariate and herd as a random effect. Calving disorder = difficult calving requiring assistance, stillbirth, or twin birth; Hyperketonemia (HYK) = blood β -hydroxybutyrate ≥ 0.7 mM at 11 ± 2 DIM; Endometritis = $\geq 11.5\%$ polymorphonuclear cells in endometrial cytology at 35 ± 3 DIM; Lameness at 49 ± 3 DIM; Clinical disease = any of retained placenta, metritis, purulent vaginal discharge, milk fever, displaced abomasum, mastitis, or lameness.

lar development (Cheong et al., 2017). However, those studies also observed an increase in circulating Hp from the prepartum baseline to 3 DIM in healthy cows. Furthermore, a robust uterine inflammatory response at calving ($\geq 35\%$ vs. $< 35\%$ PMN in uterine flushing) was associated with enhanced follicular development in the first postpartum cycle (Cheong et al., 2017). Therefore, it seems that a transient inflammatory response in the uterus immediately after parturition is part of normal immune activation and tissue repair. Conversely, we found that cows with PVD at 35 DIM or with a clinical or a uterine disease had lesser P4 concentrations after AI than cows without. Similarly, Ribeiro et al. (2016) observed lower plasma P4 concentrations at d 7 and 15 after AI in cows that had been diagnosed with uterine or nonuterine clinical disease before AI. Strüve et al. (2013) found that cows diagnosed with metritis had smaller CL after the first postpartum ovulation than cows that had not had metritis, although they had similar P4 concentrations. Similar P4 concentrations after AI were also reported by Molina-Coto et al. (2020) in cows with or without uterine inflammation assessed from 30 to 73 DIM. The methods to assess luteal function and the timing of P4 measurements varied among these studies. Nonetheless, our findings contribute to the understanding that cows experiencing peripartum health disorders, including hypocalcemia and uterine disease, have lesser P4 concentrations during early diestrus after first AI, although the mechanisms need to be investigated further.

It is uncertain if the small differences in P4 concentrations – about 0.5 ng/mL – between cows with or without health disorders are biologically important. We observed a weak relationship between serum P4 at d 8 after AI and expression of *RTP4* in multiparous cows. Carvalho et al. (2017) reported lesser *ISG15* expression (~ 5.5 vs. 7.0 relative fold change) at d 20 after AI in pregnant cows after reducing P4 by ~ 0.8 to 1.0 ng/mL on d 8 and 11 after AI using exogenous PGF_{2a} at d 5. They also observed lesser PSPB from d 32 to 67 after AI and a smaller embryonic crown-rump length at d 46 in cows with reduced P4. Conversely, supplementation of P4 with an intravaginal device inserted on d 3 after estrus and retained until embryo flushing on d 13 or 16 increased circulating P4 on d 6 by ~ 0.7 ng/mL, and supplemented heifers had larger embryos on d 13 or 16 than controls (Carter et al., 2008). Another study reported enhanced expression of endometrial genes associated with histotroph composition (i.e., triglyceride synthesis and glucose transport) in cows supplemented with a P4 intravaginal device on d 3 after AI compared with control cows (Forde et al., 2009). Finally, in a larger study with lactating cows, P4 supplementation from d 4 to 18 after AI increased plasma P4 concentrations by ~ 1.5 to 2.0 ng/mL from d 8 to d 11 (Monteiro et al., 2014). Although supplementation did not affect relative expression of *ISG15* or *RTP4* in PBL on d 19 in pregnant cows in that study, it resulted in greater pregnancy per AI in cows receiving synchronization and timed AI, but not in cows bred after detected es-

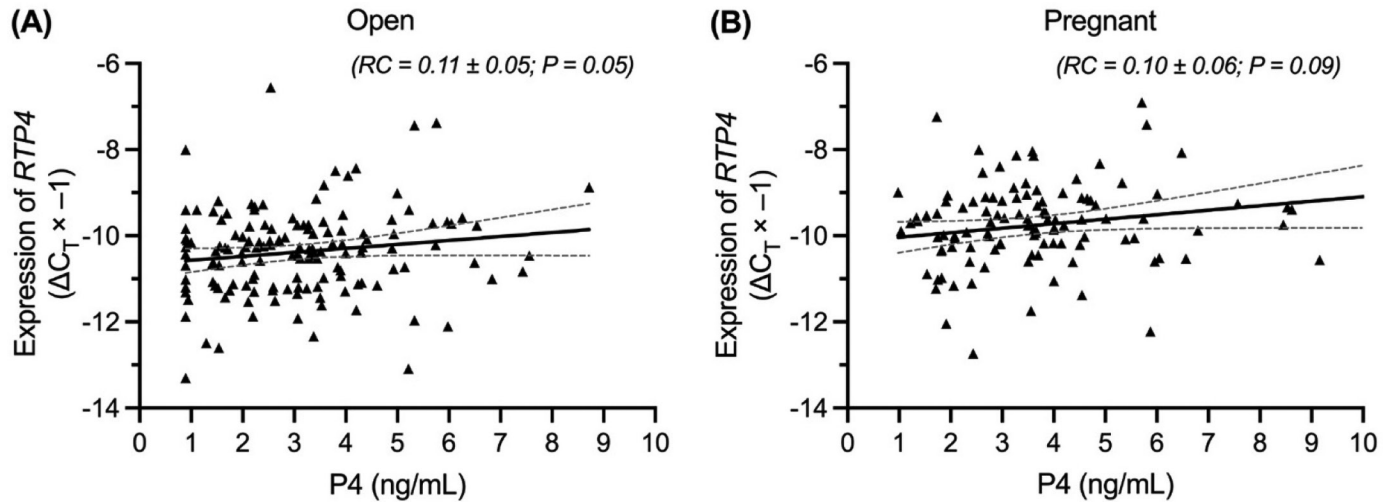


Figure 5. Association of progesterone (P4) concentration in serum on d 8 after first artificial insemination with expression of *RTP4* (on the additive inverse scale, for easier interpretation; $\Delta C_T \times -1$) in peripheral blood leukocytes on d 19 in multiparous Holstein cows from 2 herds subsequently diagnosed open (panel A; $n = 139$) or pregnant (panel B; $n = 111$) based on pregnancy-associated glycoprotein test on d 29. Regression coefficients (RC) were derived from a multivariable model in which there was an interaction among P4, parity, and pregnancy outcome in their association with *RPT4*, accounting for milk yield and displaced abomasum as covariates and herd as a random effect. The solid line is the best linear fit and the dashed lines are the 95% confidence limits around it.

trus. The associations here were of low magnitude but support the hypothesis that the effects of postpartum disease on fertility months later may be attributed in part to P4 concentrations during early to mid-diestrus after first AI.

We hypothesized that cows with postpartum disease would have reduced expression of ISG in PBL at d 19 compared with unaffected cows, as reported for cows synchronized for timed AI (Ribeiro et al., 2016). Similar to our findings of lesser *ISG15* expression in open cows previously diagnosed with HYK or ENDO, Molina-Coto et al. (2020) reported that open cows previously diagnosed with ENDO had lesser *ISG15* expression than healthy cows. In a separate analysis from this data set, we found that 16% of cows open at d 29 that had ENDO at 35 DIM were classified pregnant at d 19 based on expression of *ISG15*, compared with 35% of open cows at d 29 that did not have ENDO. This greater proportion of cows presumed pregnant at d 19 among those that did not have ENDO indicates greater relative expression of *ISG15* than those with ENDO. It is plausible that among cows open at d 29, a greater proportion of healthy than unhealthy cows had an embryo secreting IFN- τ (resulting in greater ISG expression), but they subsequently had pregnancy loss and were open at d 29. Detailed analyses on risk factors for early pregnancy losses in the present data set were reported elsewhere (Bruinje et al., 2023a). To account for these potential early pregnancy losses and to minimize the variability in ISG data between open and

pregnant cows, we built separate models that either included only cows classified as pregnant at d 19 based on *ISG15* or cows classified pregnant at d 29 based on PAG, irrespective of their status at d 19. *ISG15* was chosen over *RPT4* due to its greater accuracy in classifying pregnancy status (Table 3). Variables associated with lesser *ISG15* expression in open cows (HYK and ENDO), in addition to calving-related problems, were also associated with lesser *ISG15* expression in pregnant cows, suggesting that these health disorders contribute to pregnancy losses before d 29.

We assessed the expression of *ISG15* and *RTP4* due to their expected response to the presence of a conceptus. Previous studies demonstrated different responses of these genes to other stimuli such as intravaginal P4 supplementation after AI (Monteiro et al., 2014), recombinant bovine somatotropin treatment (Ribeiro et al., 2014), or supplementation of trace minerals of different sources (Mion et al., 2023). This suggests that different ISG could have different expression sensitivities under different conditions, such as exposure to disease in the present study. We observed lesser *RTP4* expression in pregnant cows that had a DA but greater *ISG15* expression in cows that had lameness or at least one clinical disease. In a previous study, pregnant cows that had ENDO had greater *ISG15* expression than cows that were healthy (Molina-Coto et al., 2020). Although not fully understood, there are possibly factors unrelated to pregnancy that could result in variations in ISG expression in PBL. Pregnant heifers inocu-

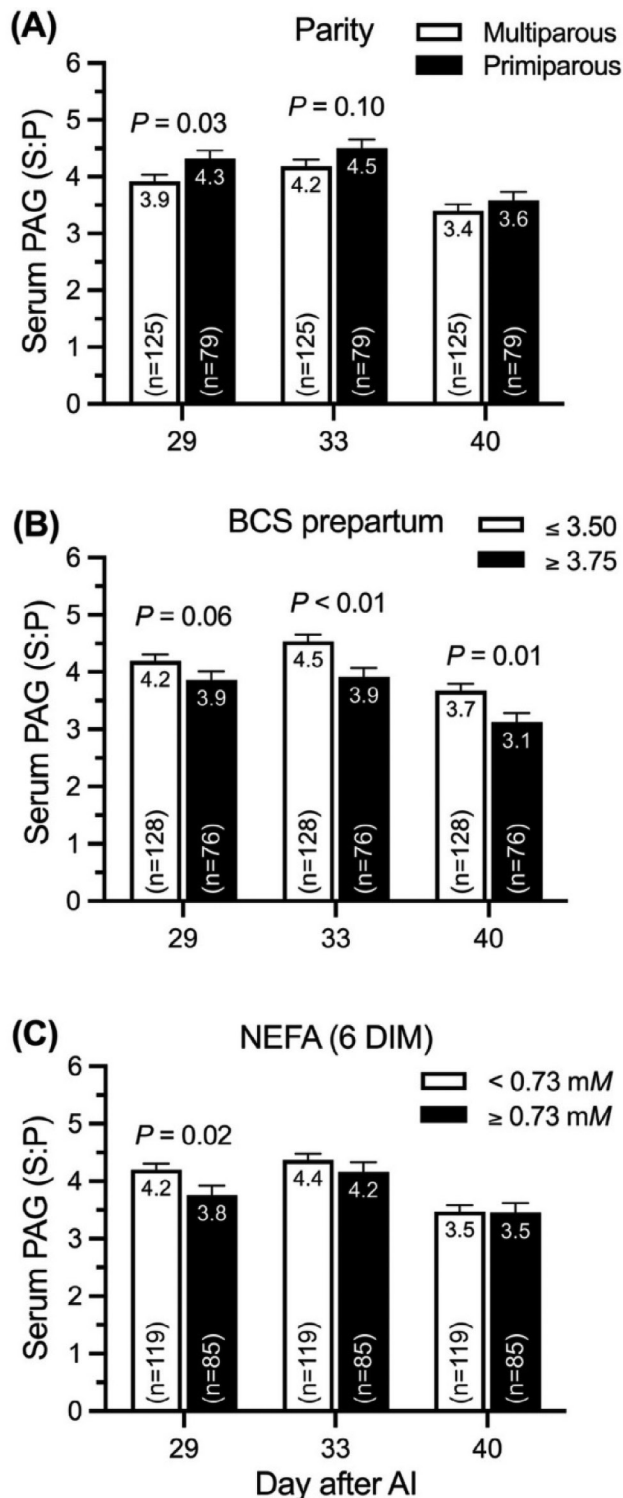


Figure 6. Associations of parity, prepartum body condition score (BCS), and nonesterified fatty acids (NEFA) concentration in serum at 6 ± 2 DIM with pregnancy-associated glycoprotein [PAG; sample to positive ratio (S:P)] measured at d 29, 33, and 40 after first postpartum artificial insemination (AI) in Holstein cows from 2 herds confirmed pregnant based on ultrasound at d 33 (* $P \leq 0.05$; # $0.05 \leq P \leq 0.10$). Results are from a multivariable model accounting for herd as a random effect.

lated with bovine viral diarrhea virus on d 75 or 175 of gestation had an upregulation of *ISG15* in whole blood in the following 2 wk, which was accompanied by pronounced antiviral activity of type-I IFN in blood (Smirnova et al., 2008). Similarly, Takino et al. (2016) reported greater *ISG15* expression in pregnant cows infected with Borna virus-1 than in non-infected pregnant cows. The authors suggested that greater *ISG15* expression in cows affected by health disorders could indicate a greater requirement of IFN- τ for pregnancy establishment signaling, or a greater sensitivity of immune cells to IFN- τ , compared with healthy cows. We did not assess viral diseases and all cows in these herds were negative for bovine viral diarrhea virus. However, we speculate that immune activation, possibly linked to lameness that was often diagnosed close to time of AI (49 DIM), could be linked to altered expression of ISG in PBL. Further research should investigate if disease processes, whether associated with viral infection or not, can affect the expression of ISG in PBL.

As markers of metabolic disorders, HYK and DA were associated with ISG expression, and *ISG15* expression was weakly but positively associated with PAG levels. Therefore, metabolic disorders potentially affecting IFN- τ secretion might also affect placental function, though that requires further investigation. In contrast to Mercadante et al. (2016), we did not find associations of uterine or clinical disorders with PAG levels. In a separate analysis from the same data set (Bruinje et al., 2023a), we demonstrated that cows with uterine diseases (metritis, ENDO, or PVD) were less likely to be pregnant on d 19 (based on *ISG15*) and on d 29, 33, or 40 (based on PAG), as well as more likely to undergo a pregnancy loss from d 19 to 40. We speculate that uterine disorders mainly affect early pregnancy establishment, but not placental function as reflected by PAG levels, at least among pregnancies that survived to d 40.

This study was conducted on 2 commercial farms that had management protocols which could potentially interfere with our conclusions. For instance, cows were treated for HYK and lameness (2 variables associated with ISG expression) based on different criteria than used in our models. Cows received treatment for HYK based on a higher threshold (1.2 mM), and for lameness only if they were visibly or severely lame and detected by the farm staff, which was independent of our assessment of lameness. If treatment mitigated potential effects of HYK or lameness on the outcomes evaluated, it would have been applied only to a portion of cows (53% of the cows with HYK at 8 DIM, 20% of the cows with HYK at 11 DIM, and 34% of the cows with lameness). There were too few cows in these subcategories for valid analysis with our data. Separately, most cows

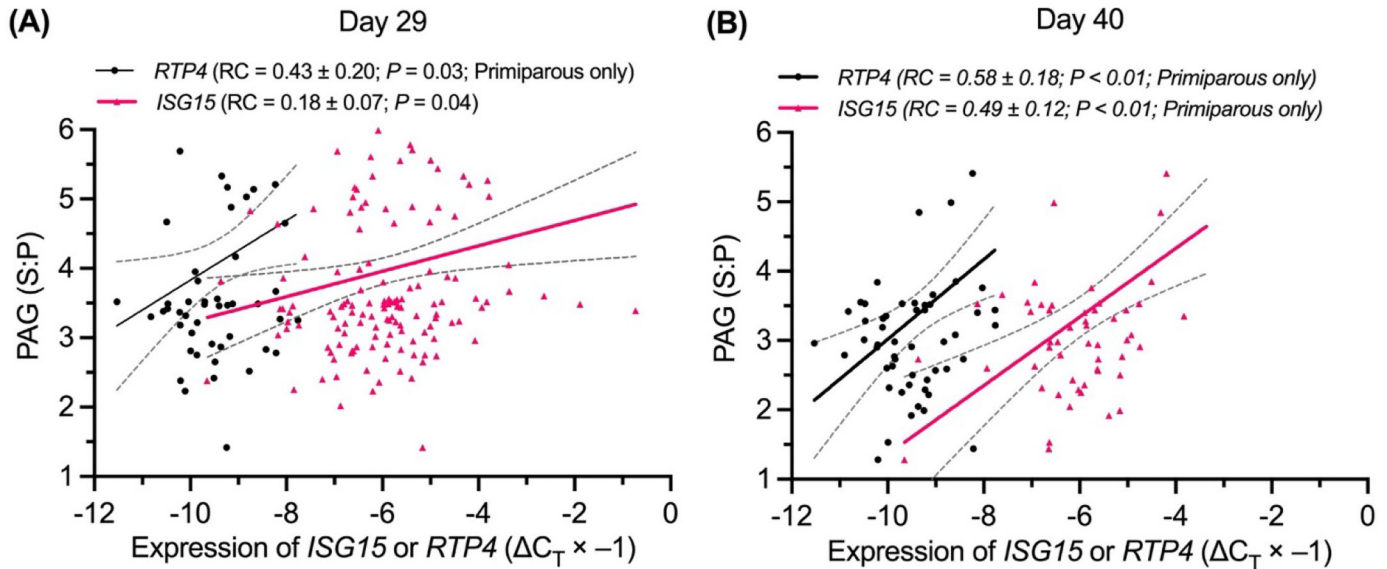


Figure 7. Associations of expression of interferon-stimulated genes (*ISG15* and *RPT4* on the additive inverse scale, for easier interpretation; $\Delta C_T \times -1$) in peripheral blood leukocytes on d 19 after first artificial insemination (AI) with pregnancy-associated glycoprotein (PAG) measured in serum on d 29 (panel A) or 33 (panel B) after AI in Holstein cows from 2 herds diagnosed pregnant based on PAG test on d 29. There was an interaction between *RPT4* expression and parity on their association with PAG S:P on d 29 ($P = 0.07$) and on d 40 ($P = 0.03$), and interaction between *ISG15* expression and parity on their association with PAG S:P on d 40. Regression coefficients (RC) were derived from multivariable models that accounted for the covariates of parity, body condition score, and DIM at AI, with herd as a random effect (for *ISG15* in panel A; $n = 154$), or for DIM at AI as covariate and herd as a random effect for the remained models (primiparous cows only; $n = 59$). The solid lines are the best linear fit and the dashed lines are the 95% confidence limits around it.

were bred at estrus detection between 50 and 75 DIM, with those not detected in estrus bred later with timed AI. Although DIM and method of AI were evaluated as covariates, this limits comparison of our findings with studies that synchronized cows for timed AI at similar DIM.

CONCLUSION

Our findings suggest that postpartum health disorders, particularly metabolic disorders associated with energy status or reproductive tract disease, can have small carryover effects on luteal function during early diestrus after first AI, pregnancy recognition signaling, and placental function during early pregnancy. Most of the associations identified here between postpartum health disorders and reproductive function variables were small in magnitude, and some were directionally opposite to our hypothesis. However, these associations may partially explain some of the pathways linking postpartum health and pregnancy establishment and maintenance.

ACKNOWLEDGMENTS

This study was supported by the Ontario Ministry of Agriculture, Food and Rural Affairs and the Natural

Sciences and Engineering Research Council of Canada (NSERC), and by doctoral scholarships provided by Dairy Farmers of Ontario and NSERC. We thank IDEXX Laboratories Inc. for donating the BVDV and Bovine Pregnancy Test kits, and Drs. Julio Giordano (Cornell University) and Trevor DeVries (University of Guelph) for providing valuable comments in the preparation of this manuscript. We also thank the dairy producers for their cooperation and willingness to participate in the study and the students and research technicians who assisted with data collection.

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